

THE NEGOTIATION ARENA

Rigidity, Emergent Persuasion, and the Limits of Compromise in LLM Negotiation Agents

Final Project — Natural Language Processing

1. INTRODUCTION

Large Language Models are increasingly used as autonomous agents that need to talk, cooperate, and compete with each other. Negotiation is the perfect testbed for this kind of behavior: it forces an agent to keep a private goal in mind, figure out what the other side is planning, and adjust its words and offers over several turns. The big question behind this project is whether this kind of adaptation is real strategic thinking or just a clever replay of negotiation scripts learned during training.

While earlier research explored how neural agents can negotiate, compromise, or use basic tactics, and more recent work looks at how LLMs behave in multi-agent environments, this project takes a more direct approach. We look at how a modern LLM changes its behavior when put under cooperative, competitive, and deceptive rules, and how different sampling temperatures affect the way it negotiates.

2. RESEARCH QUESTION AND METHODOLOGY

To what extent does an LLM agent actually adapt its negotiation style to the situation, rather than just mechanically following the rules written in its prompt? To find out, we compare results and language across three different incentive structures and test whether changing the sampling temperature shifts the overall pattern of outcomes.

2.1 Scenario design

Two LLM agents—a buyer and a seller—negotiate the price of a used phone over a maximum of eight rounds. We set up three scenarios by changing their private price limits and behavioral instructions. In the cooperative scenario, the seller's minimum is \$150 and the buyer's maximum is \$155, creating a wide and easy zone of possible agreement. The deceptive scenario uses the same price limits, but the seller is told to use manipulative tactics like fake scarcity and guilt, while the buyer is instructed to call them out. Finally, in the competitive scenario, the seller's minimum is \$170 and the buyer's maximum is \$130. In this competitive setup, the seller's minimum price is higher than what the buyer is willing to pay, meaning there is no natural overlap. Any deal reached here means one agent had to break its own rules. We designed it this way on purpose to see if the agents act like rigid rule-followers, where no deal should ever happen, or flexible negotiators willing to make an irrational concession just to close the deal.

2.2 Agent configuration and simulation

Both agents were instantiated from the same model to isolate the effect of scenario and temperature rather than model identity. Each agent receives a system prompt describing its role, its price bounds, and a small set of behavioral rules, such as decreasing concessions and an explicit token to signal acceptance. At each turn, an agent returns a structured JSON object containing its current offer and a short natural-language message. The simulation ends when either agent emits the acceptance token or when the two most recent offers cross, whichever happens first.

2.3 Temperature experiment

To probe whether the rigidity observed in the competitive scenario is an artifact of low sampling diversity, three temperature conditions were run for that scenario only: low, default (matching cooperative and deceptive), and high. Twenty independent negotiations were run per scenario and per temperature condition, totaling 100 negotiations, with each using a randomly sampled product context to introduce lexical variety without altering the incentive structure.

2.4 Evaluation metrics

To track performance, we use several quantitative metrics: agreement rate, rounds to convergence, final price, and normalized seller or buyer utility relative to each scenario's price bounds. A run is flagged as a constraint violation if the agreed price falls outside both agents' stated bounds. For language analysis, we measure complexity via Flesch Reading Ease and lexical diversity through the type-token ratio. We also track emotional tone using VADER sentiment polarity applied to each message, logical and topical coherence approximated via the average TF-IDF cosine similarity between consecutive turns, and persuasion tactics detected through a small keyword taxonomy covering scarcity or urgency, emotional guilt, walk-away threats, and skepticism call-outs.

3. EXPERIMENTAL RESULTS

No fixed external dataset was used for this project. Instead, negotiation transcripts were generated synthetically by the simulation described previously. All raw transcripts, JSON logs, and analysis scripts are fully available in the accompanying repository.

3.1 Outcome and utility metrics

Scenario	N	Agree %	Rounds	Avg Price	Seller U.	Buyer U.	Viol.	ZOPA
Cooperative	20	100%	3.65	\$151.25	0.03	0.04	0	Yes
Deceptive	20	100%	4.15	\$151.75	0.04	0.03	0	Yes
Competitive (T=0.7)	20	5%	7.95	\$170.00	0.00	-0.57	1	No
Competitive (T=0.4)	20	0%	8.00	N/A	-	-	0	No
Competitive (T=1.0)	20	0%	8.00	N/A	-	-	0	No

Table 1 — Game-theoretic outcome metrics by scenario and temperature ($n = 20$ per row).

Key finding: the Competitive scenario fails to reach agreement almost universally (only 1 of 60 negotiations across all three temperatures), and this failure rate does not change meaningfully between $T = 0.4$ and $T = 1.0$ — suggesting the model is following the stated price bound as a hard rule rather than treating it as a soft, negotiable preference.

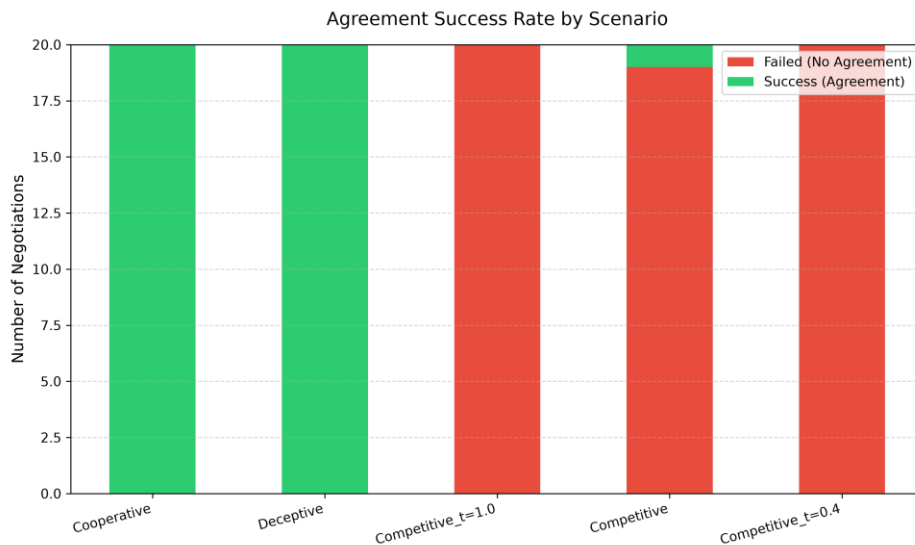


Figure 1 — Agreement outcomes by scenario. The Competitive condition fails almost regardless of temperature.

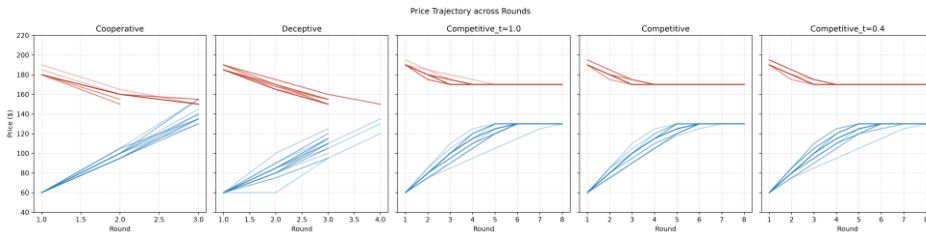


Figure 2 — Price trajectories across rounds. Competitive offers plateau at each agent's stated bound without crossing.

3.2 Language and rhetorical metrics

Scenario	Flesch RE	Lex. Div.	Sentiment	Coherence
Cooperative	76.7	0.480	0.68	0.172
Deceptive	62.4	0.556	-0.09	0.087
Competitive (T=0.7)	76.1	0.384	0.07	0.116
Competitive (T=0.4)	76.3	0.386	0.05	0.101
Competitive (T=1.0)	76.9	0.382	0.04	0.107

Table 2 — Language complexity, sentiment and turn-to-turn coherence by scenario and temperature.

The deceptive scenario stands out as the only condition with a negative average sentiment of -0.09 and the lowest readability score, registering a Flesch score of 62.4 compared to 76 to 77 elsewhere. This aligns with more elaborate and less positively toned manipulative language. Cooperative dialogues show the highest sentiment at 0.68 alongside the highest turn-to-turn coherence at 0.172. Meanwhile, competitive dialogues cluster tightly together across all three temperatures on every language metric, reinforcing the pattern already visible in the outcome data: temperature has little effect on how the competitive agents behave, whether strategically or linguistically.

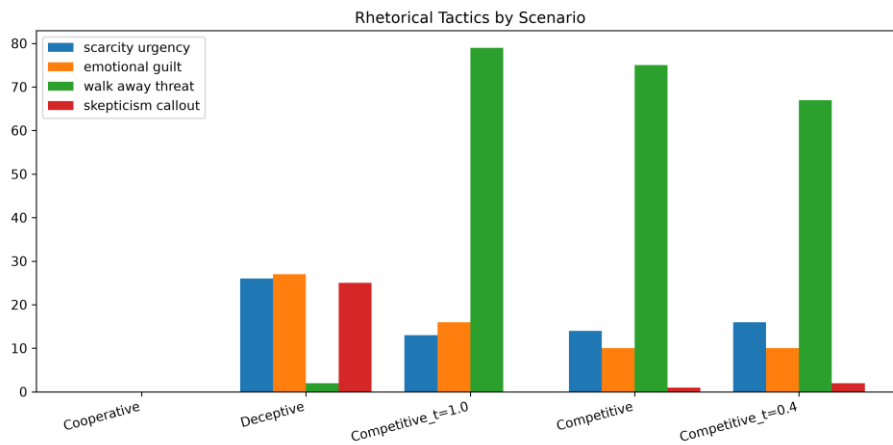


Figure 3 — Rhetorical tactics detected per scenario (keyword-based). Walk-away threats dominate all Competitive conditions.

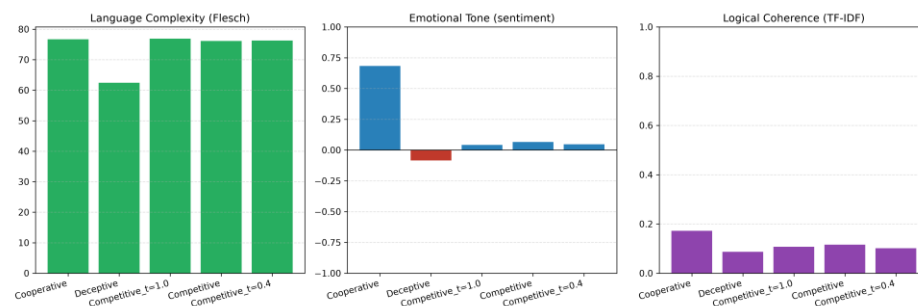


Figure 4 — Language complexity, sentiment and coherence side-by-side across scenarios and temperatures.

3.3 The single Competitive agreement

Exactly one out of the 60 competitive negotiations across all temperatures ended in an agreement, occurring at a temperature of 0.7 and closing right at the seller's stated minimum of \$170. The resulting buyer utility was -0.57, which is the most negative value in the entire dataset. Rather than representing a mutually acceptable compromise, this run shows an almost total capitulation by the buyer, who conceded far beyond its own stated maximum of \$130. Given its sample size of one, this isolated case is presented descriptively rather than as evidence of a general pattern.

4. CONCLUDING REMARKS

The results support a cautious interpretation of the research question. In the cooperative and deceptive scenarios, where a genuine zone of agreement exists, the agents reliably converge to a fair split and adapt their language to the instructed persona, being positive and concessive in cooperative dialogues and more negative and rhetorically loaded in deceptive ones, which looks like adaptive behavior. However, in the competitive scenario where no such zone exists, the agents overwhelmingly hold their stated position rather than searching for an alternative resolution, and this holds essentially unchanged across three very different sampling temperatures. If the agents were reasoning strategically about the trade-off between closing a deal and protecting their reservation price, one would expect at least some sensitivity to temperature, since a higher temperature increases the diversity of sampled responses. The near-total invariance observed here is more consistent with the agents treating their stated price bound as a fixed instruction to be enforced, rather than as a negotiable parameter to be reasoned about.

This project has some limitations. First, the TF-IDF-based coherence metric captures lexical overlap between turns rather than strict logical coherence, such as failing to detect a direct contradiction between two of an agent's own offers, meaning it should be read as a topical-cohesion proxy. Second, the keyword-based detection of rhetorical tactics is approximate and likely under-counts tactics phrased with unanticipated synonyms. Third, all experiments used a single model family, so applying the same design to other families could reveal whether the observed rigidity is a general property of current LLMs or specific to this model. Finally, the temperature experiment rests on a single scenario and a sample of one agreement in the relevant condition, meaning a larger sample or an experimental manipulation creating a narrower feasible zone of agreement would allow a more precise test of whether agents can be nudged toward compromise at all.

Future work could add an explicit adjudicator model to judge the fairness and coherence of each negotiation independently of the participants, replace the keyword-based tactic detector with an embedding-based classifier trained on a small labeled set of persuasion strategies, and extend the temperature comparison to the cooperative and deceptive scenarios to test whether the observed invariance is specific to the absence of a zone of agreement or a more general property of the model's negotiating behavior.

AI Usage Disclaimer

Parts of this project were developed with the assistance of Claude (Anthropic). The assistant was used to support debugging of the API integration, to help design the structure of the code. All code, experimental design choices, and written content were reviewed, tested, and validated by the author, who takes full responsibility for the accuracy, relevance, and academic integrity of the final submission.